

CPSC 304 Project Cover Page

Milestone #: 1

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Group Number: 25

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia.

Deliverables

2.

a. **What is the domain of the application? Describe it**

The domain of this application is ski resort management. It encompasses the administration and coordination of various aspects of a ski resort such as ski runs and lift operations, equipment rentals, ski lessons, dining reservations, and passes. The goal is to provide a system that organizes and optimizes the operations of a ski resort.

b. **What aspects of the domain are modeled by the database?**

The database models several important aspects of ski resort management to address the needs of both guests and the resort. This includes:

- **Guest Management:** The database stores information about patrons, including details such as their names, addresses, contact information, and reservation history. This models the interactions guests have with various services offered by the resort, as patrons can *rent* equipment, *buy* lift tickets, *take* classes, and *eat* at restaurants.
- **Equipment Rental:** The Rental entity is connected to Patron through the *rents* relationship. This models the process of guests renting ski equipment and helps track rental transactions and inventory. It is a weak entity that depends on the Patron entity, represented through the *rents* relationship as rental records cannot exist without an associated guest.
- **Lift Ticket Management:** The Lift Ticket entity has an ISA relationship with Day Pass and Season Pass, showing different types of passes available. Patrons can *buy* lift tickets, which are necessary to access ski lifts and runs. This models the ticketing system and helps in tracking ticket sales and usage.
- **Ski Lifts and Runs:** The Ski lift entity, connected through the *rides* relationship, and Ski run entity, connected through the *access* relationship, models the infrastructure of the resort. These relationships indicate which ski lifts provide access to which ski runs and which classes ride which lifts.
- **Ski Instruction (Classes):** The Class entity, has a relationship *teaches* to Ski instructor and *takes* to Patron, which models the ski lessons offered by the resort. This helps in scheduling and managing classes, as well as tracking which patrons are taking lessons from which instructors.
- **Dining Services:** The Restaurant entity, connected through the *eats* at relationship with Patron, models the dining services available at the resort. The discount relationship indicates that patrons with lift tickets may receive discounts at restaurants. This aspect helps in managing dining reservations and offering promotional deals to guests who wish to purchase food.

3.

a. What functionality will the database provide?

This database can be used by ski resort management to keep track of the status of all parts of the mountain. Ski instructors can use the lift and class information to see what parts of the mountain are open as well as what runs have many classes running on them to plan out where they will take their class on the mountain. Upper management can use the information regarding patrons, passes, and dining locations to drive future pricing and improvements to the resort. Maintenance crews can use lift and run status to see where maintenance is needed on the mountain and help reduce downtime.

4.

a. What database will your project use?

We will use the department provided Oracle database.

b. What is your expected application technology stack?

We will be using NodeJS with the Express.js framework and oracledb module.

